Sivan Syed

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Education

The University of Texas at Austin (GPA: 3.8/4.0) **May 2027**
B.S. ***Electrical Engineering****, B.S. Physics, B.S. Mathematics, Certificate German*
Relevant Coursework: Embedded Systems, Solid-State Devices, Power Electronics, Fundamental Electronic Circuits
Graduate Coursework: Magnetic Materials and Devices, Computational Physics, Quantum Mechanics I

Experience

Emerson, Electrical Engineering Design Intern **May 2026–Aug 2026**
- Captured a 4–20 mA op-amp and potentiometer signal-conditioning schematic enabling field calibration of set-points
- Reverse-engineered a raw production PCB to author complete schematics and assembly Work Instructions

Adom Industries Inc., R&D Design Engineer **May 2025–Aug 2025**
- Architected a 48V VCU (RP5+RP2040) for ROS path planning via CAN and a hardware E-stop and isolated 5V logic rail
- Designed custom PCBs for LiDAR and motor control, applying digital isolators to suppress 48V EMI

Sandia National Laboratories, R&D Computational Engineer **May 2024–Aug 2024**
- Built a fast current-pulse generator modeling capacitor discharges, synthesizing 10k+ waveforms for ML surrogate modeling and Uncertainty Quantification
- Analyzed fluid RTIs and RMIs to diagnose simulation inaccuracies
- Wrote ODE trajectory solvers to track radial liner implosions and compare observed instability spectra

Banerjee Lab – 2D Material Synthesis & Characterization, Undergraduate Researcher **Sep 2025–Present**
- Characterized and grew MoS_2 monolayers via CVD to optimize carrier type, doping, and bandgap
- Employed XRD, Raman, and Photoluminescence spectroscopy for material analysis

FSAE Longhorn Racing, Dynamics + Vehicle Modeling Engineer, Steering Lead **Sep 2022–Aug 2025**
- Modeled vehicle dynamics in a Model-in-the-Loop simulation, validating slip-angle targets against track telemetry

Projects

Safety-Critical UAV Flight Controller
- Building a custom 4-layer PCB running a bare-metal control loop and TinyML anomaly detector under FreeRTOS
- Minimizing ESC-induced EMI with dedicated ground planes, digital isolators, and star-grounded LC pi-filters

Embedded Racing Simulator
- Implemented multi-node CAN bus network, debugging bus arbitration and extending the data payload to support higher-bandwidth real-time telemetry between the steering wheel, pedal box, and central compute
- Designed custom PCBs in KiCAD for steering wheel-mounted TM4Cs, integrating CAN transceivers and signal conditioning for a rotary encoder–based throttle/brake input

High-Power E-Bike Boost Converter
- Engineered a custom DC-DC boost converter PCB, suppressing radiated EMI by minimizing the high di/dt critical loop and gate-drive parasitic inductance
- Architected a digital control loop on a TI C2000 MCU, synchronizing the ePWM module with the ADC via hardware triggers to sample feedback free of switching noise

SOT-Driven Skyrmion Racetrack Memory Device
- Designed a novel SOT-driven magnetic skyrmion racetrack memory in MuMax3, using a passive defect channel and VCMA gate to pin, release, and steer skyrmions on demand
- Automated a 33-job micromagnetic sweep on TACC Lonestar6 (A100 GPUs) via SLURM, diagnosing device failure mode through topological-charge and velocity analysis

Skills

Languages: C, C++, ARM, Rust, Python, MATLAB, Bash, JS, HTML, CSS, LaTeX
Tools: KiCAD, Cadence, Draftsight, LTSpice, MuMax3, Fusion360, SOLIDWORKS, Ansys